2. **Which of the following latitudes (north or south) would you associate with 'Horse Latitudes' ?**

(a) 30° (b) 45°

(c) 60° (d) $23\frac{1^0}{2}$

U.P. Lower (Spl) (Pre) 2008

Ans. (a)

The horse latitudes are located at about 30° to 35° north and south of the Equator. In the northern hemisphere it is called the north subtropical high-pressure belt and in the southern hemisphere, it is known as the south subtropical high-pressure belt. The existence of these pressure belts is due to the fact that the rising air of the equatorial region is deflected towards poles due to the earth's rotation. After becoming cold and heavy, it descends in these regions due to Earth rotation and gets piled up. This results in high pressure. Calm conditions with feeble and variable winds are found here. In ancient times vessels with a cargo of horses passing through these belts found difficulty in sailing under these calm conditions. They used to throw a few horses in to the sea in order to make the vessels lighter. Thus, these belts or latitudes came to be known as 'horse latitudes'.

3. **Consider the following statements:**

1. **Either of the two belts over the oceans at about 30^o to 35^o N and S latitudes is known as Horse Latitudes.**
2. **Horse Latitudes are low pressure belts.**

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2007

Ans. (a)

See the explanation of the above question.

4. **Air pressure is lowest in :**

(a) Winter season (b) Spring season

(c) Autumn season (d) Summer season

U.P.P.C.S. (Pre) 2014

Ans. (d)

Air pressure is defined as the pressure exerted by the weight of air per unit surface area on the earth's surface. The distribution of air pressure is influenced by the temperature of air at a given place. Temperature is inversely related pressure. Due to highest temperature in summer, pressure is lowest on earth. However, as altitude increases temperature and air pressure decrease. Distribution of air pressure is influenced by temperature and other kinetic factors (i.e. earth rotation, water vapor, altitude) etc.

Clouds

∗A cloud is a visible dense mass of water droplets or ice crystals suspended in the air. Hence, collectively clouds are a large version of suspended water bodies or glacial bodies. Clouds are found at a relatively higher altitude compared to fog.

∗Clouds can be categorized depending on their shape, vertical expansion and height.

Cirrus Clouds – They clouds are found at the highest altitude. These are fibrous and feathery in appearance as they are made up of tiny ice crystals. When the rays of the sun pass through these ice crystals the cirrus clouds appear white in colour. They are seen before the advent of a cyclone.

Cirro Stratus Cloud – They are found at an altitude of 18000 feet and above. Their arrival creates a halo around the Sun and the Moon which is a sign of an upcoming cyclone.

Cirro-Cumulus Clouds – These clouds are white in appearance and are found in clusters of small ball-shaped patches.

Alto-Stratus Clouds – These are those high clouds that are spread across the sky as thin sheets of blue and brown clouds. They are responsible for continuous and expanded rainfall.

Alto – Cumulus Clouds – They are low-level clouds just like cirro-cumulus clouds.

Strato-Cumulus-Clouds – These are light brown coloured clouds often found in big round shaped patches.

Stratus Clouds – They are formed as a result of two opposite air bodies, coming into contact with each other. They are often seen during winter in temperate regions.

Nimbo-Stratus Clouds – They are found close to the surface of the Earth. These black clouds are highly dense in nature and can be of any shape. It is because of their highly dense nature that they block the sunlight and make the region dark followed by heavy rainfall.

Cumulus Clouds – They are highly dense in nature. Their shape is elongated and the upper portion of them resembles a dome or a cauliflower. These clouds are a signal of the start of clear weather.

Cumulo-Nimbus-Clouds – These clouds are highly diversified and dense in nature. There is a high possibility of rain, hail and thunderstorms associated with these clouds.

∗Thunder is a process of a thunderstorm. In reality, thunderstorms are local storms in which the air ascends with high velocity. Lightning and thunder is witnessed followed by heavy rainfall. Lightning causes a sudden and high increase in the temperature of the air leading to a sudden expansion in the air which result in thunder.

1. Clouds are the result of –

(a) Evaporation (b) Normal temp. lapse rate
(c) Catabatic lapse rate (d) Condensation

Uttarakhand P.C.S. (Pre) 2016

Ans. (d)

Clouds are formed as a result of the process of condensation. Conversion of water in the gaseous state to a liquid or solid state is called condensation. Clouds are a collection of very tiny droplets of water or ice crystals formed by condensation of water vapor in the atmosphere.

2. Which of the following cloud is responsible for highly intense rain?

(a) Cumulus (b) Cumulonimbus
(c) Nimbostratus (d) Cirrostratus

Jharkhand P.C.S. (Pre) 2003

Ans. (b)

Cumulo-nimbus clouds are highly diversified and dense in nature. These are a high possibility of rain, hail and thunderstorms associated with these clouds. Cumulonimbus cells usually dissipate within an hour once they start falling, making for short-lived, heavy rain.

3. Highest altitude clouds are -

(a) Altocumulus
(b) Altostratus
(c) Cumulus
(d) Cirrostratus

U.P. Lower Sub. (Pre) 2009

Ans. (d)

Cirrostratus is the highest altitude cloud. These are mainly formed at an altitude of over 18000 feet.

4. Consider the following climatic and geographical phenomena:

1. Condensation
2. High temperature and humidity
3. Orography
4. Vertical wind

Thunder clouds development is due to which of these phenomena?

(a) 1 and 2 (b) 2, 3 and 4
(c) 1, 2 and 4 (d) 1, 2, 3 and 4

I.A.S. (Pre) 2002

Ans. (c)

Thunder cloud are a process of a thunderstorm. In fact, thunderstorms are a local lightning storms that develop in hot, humid tropical areas like India very frequently. The rising temperature produces strong upward winds. These winds carry water droplets upwards, where they freeze(condense), and fall down again. The swift movement of the falling water droplets along with the rising air creates lightning and sound. So it is clear that except orography, all are related to thunder clouds.

5. During a thunderstorm, the thunder in the sky is produced by the –

1. meeting of cumulonimbus clouds in the sky.
2. lightning that separates the nimbus clouds.
3. violent upward movement of air and water particles.
Select the correct answer using the code given below

(a) Only 1
(b) 2 and 3
(c) 1 and 3
(d) None of the above

I.A.S. (Pre) 2013

Ans. (d)

During a thunderstorm, thunder is the sound caused by a lightning discharge. Lightning heats the air in its path and causes a large overpressure of the air within its channel. The channel expands supersonically into the surrounding air as a shockwave and creates an acoustic signal that is heard as thunder. Hence none of the above options is correct.

6. In the context of which of the following do some scientists suggest the use of cirrus cloud thinning technique and the injection of sulphate aerosol into stratosphere?

(a) Creating the artificial rains in some regions
(b) Reducing the frequency and intensity of tropical cyclones
(c) Reducing the adverse effects of Solar Wind on the Earth
(d) Reducing the global warning

I.A.S. (Pre) 2019

Ans. (d)